www.PapaCambridge.com

O

UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

PHYSICS **5054/13**

Paper 1 Multiple Choice **May/June 2010**

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

7193734327

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

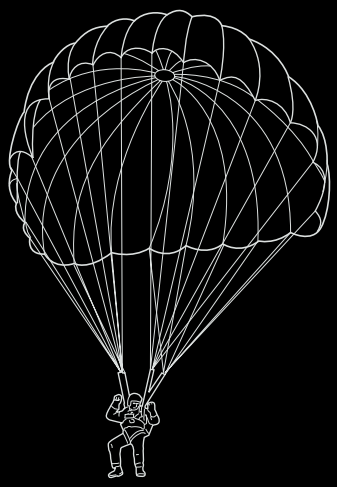
There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

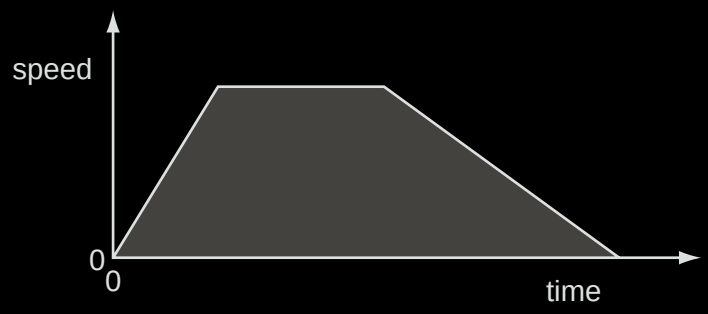
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

UNIVERSITY of CAMBRIDGE
International Examinations

www.PapaCambridge.com



www.PapaCambridge.com



What does the shaded area of the graph represent?

A the average acceleration of the car

B the average speed of the car

C the total distance travelled by the car

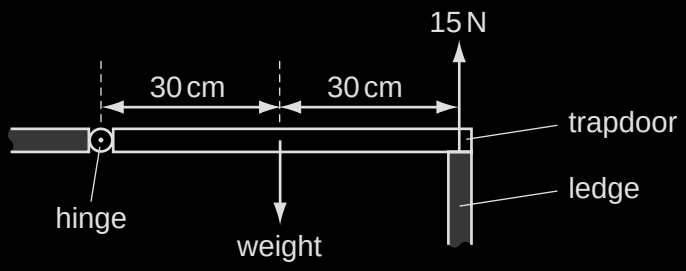
D the total travelling time of the car

6 A passenger is sitting in an aeroplane, which takes off and climbs to 10 000 m.

During this time, what happens to the mass and to the weight of the passenger?

	mass	weight
A	decreases	decreases
B	increases	increases
C	unchanged	decreases
D	unchanged	increases

www.PapaCambridge.com



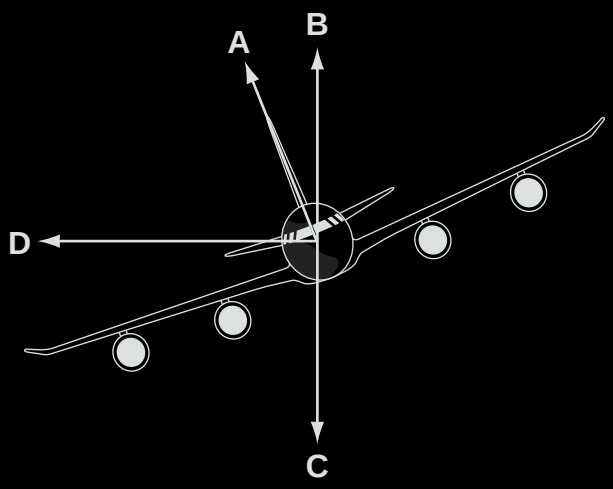
When the trapdoor is closed, the ledge exerts an upward force of 15 N on the trapdoor. The gravitational field strength is 10 N / kg.

What is the mass of the trapdoor?

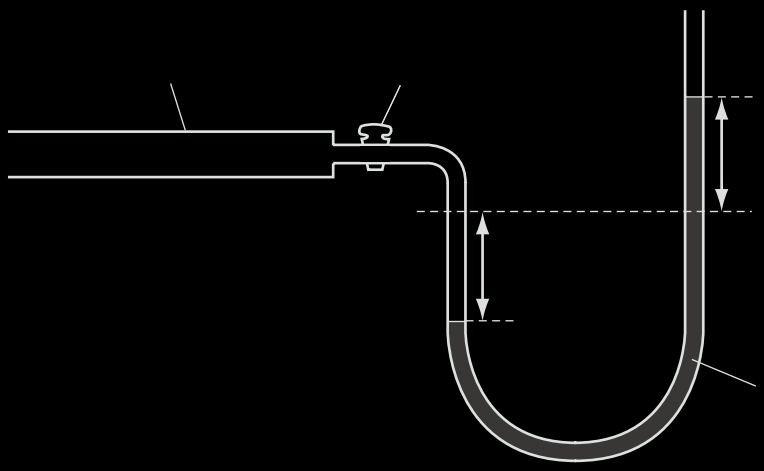
A 1.5 kg **B** 3.0 kg **C** 30 kg **D** 150 kg

8 The diagram shows an aeroplane turning in a horizontal circle at constant speed.

In which direction is there a resultant force?



www.PapaCambridge.com



What is the pressure of the gas supply?

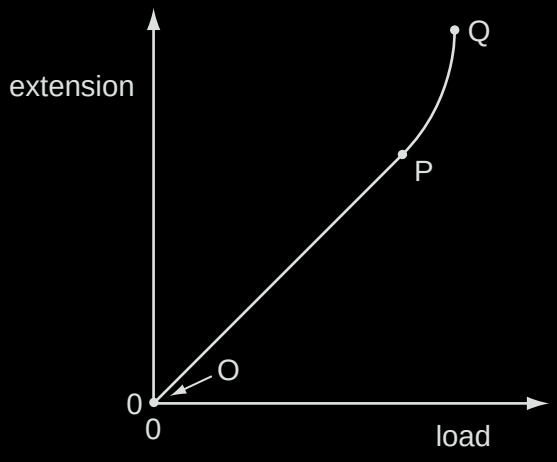
A 18 cm of liquid below atmospheric pressure

B 9 cm of liquid below atmospheric pressure

C 9 cm of liquid above atmospheric pressure

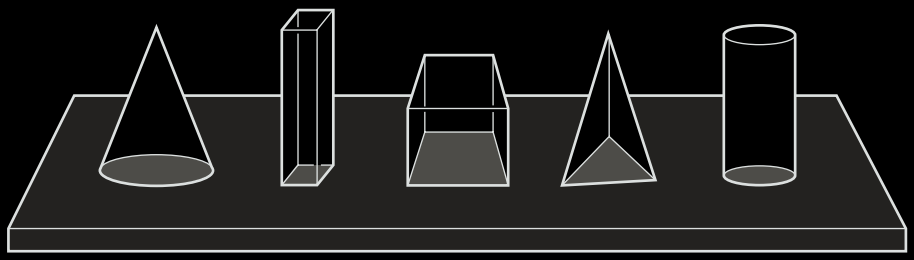
D 18 cm of liquid above atmospheric pressure

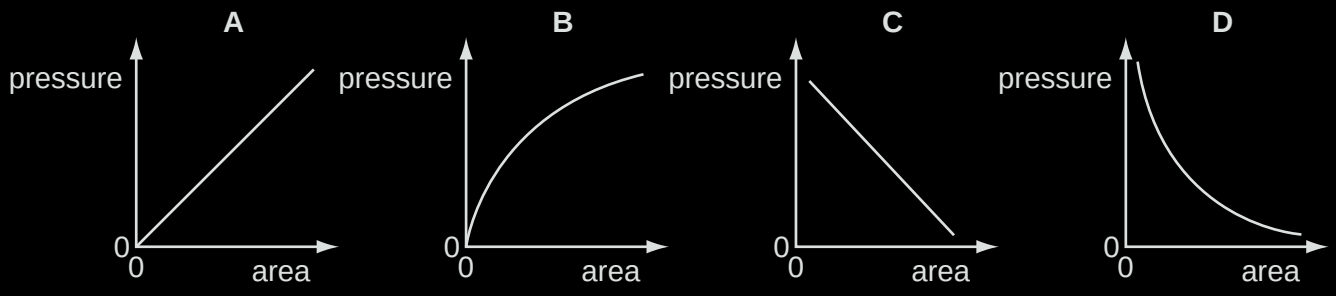
10 Which part of the graph shows the limit of proportionality for an elastic solid?



A O **B** OP **C** P **D** PQ

www.PapaCambridge.com





12 A small emergency generator supplies 432 000 000 J of electrical energy in twenty-four hours.

What is the average power output of the generator?

A 5000 W

B 300 000 W

C 18 000 000 W

D 432 000 000 W

13 A parachutist has opened his parachute and is falling to Earth at constant speed.

What is the principal energy conversion taking place as he falls?

A kinetic energy → potential energy

B kinetic energy → thermal energy (heat)

C potential energy → kinetic energy

D potential energy → thermal energy (heat)

www.PapaCambridge.com



When the strip is heated, the brass expands more than the iron.

Which shape will the strip become?

A



B



C



D



www.PapaCambridge.com



What happens to the density of this air and to its position inside the refrigerator?

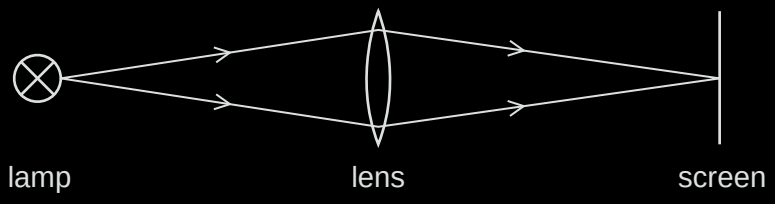
	density	position of the air
A	decreases	sinks to the bottom
B	decreases	stays at the top
C	increases	sinks to the bottom
D	increases	stays at the top

18 Convex lenses are used in cameras and as magnifying glasses.

Which types of image are formed?

	type of image in camera	type of image in magnifying glass
A	real	real
B	real	virtual
C	virtual	real
D	virtual	virtual

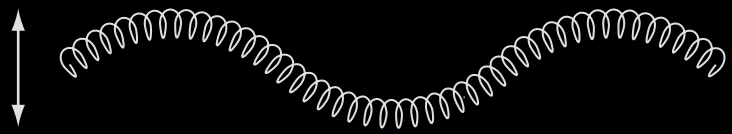
www.PapaCambridge.com



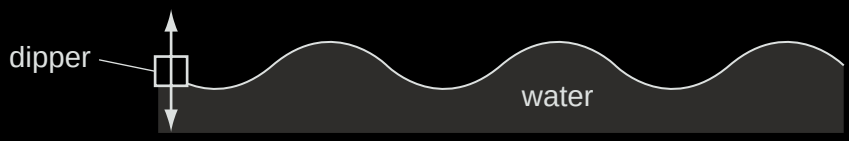
B a spring pushed backwards and forwards



C a spring pushed up and down



D a water ripple caused by a dipper moving up and down



20 A flash of lightning and the corresponding sound of the thunder are detected 6 s apart. A student calculates that the lightning struck about 1800 m away.

On which assumption is the calculation based?

A Light reaches us almost instantaneously, but sound travels at 300 m / s.

B Light travels 300 m / s faster than sound.

C Light travels 300 times faster than sound.

D The sound of the thunder was emitted 6 s after the flash.

21 A hospital needs to sterilise medical equipment.

Which electromagnetic waves could be used?

A infra-red

B microwaves

C radiowaves

D ultraviolet

www.PapaCambridge.com



A



B



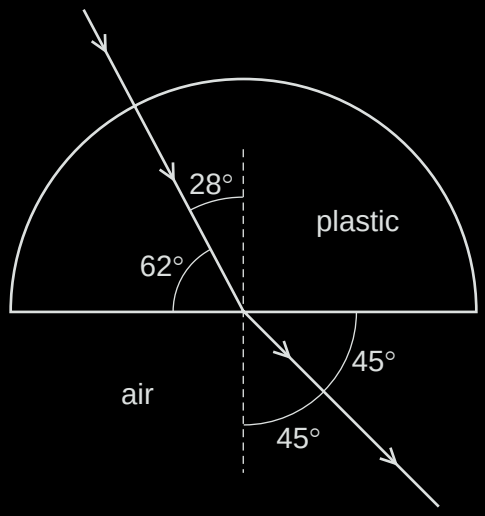
C



D



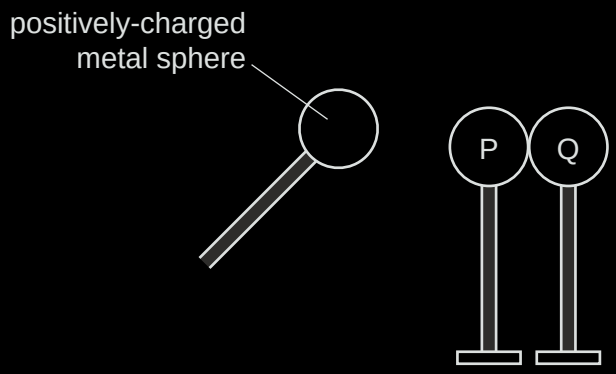
23 A semi-circular block is made from a plastic. A ray of light passes through it at the angles shown.



To two decimal places, what is the refractive index of the plastic?

A 1.25 **B** 1.41 **C** 1.51 **D** 1.61

www.PapaCambridge.com



The positively-charged metal sphere is held in this position and sphere Q is moved to the right, away from sphere P.

What are the signs of the induced charges on P and Q and how do the sizes of these charges compare?

	charge on P	charge on Q	sizes of the charges
A	negative	positive	equal
B	negative	positive	unequal
C	positive	negative	equal
D	positive	negative	unequal

25 Which properties make materials suitable for use as a core in an electromagnet?

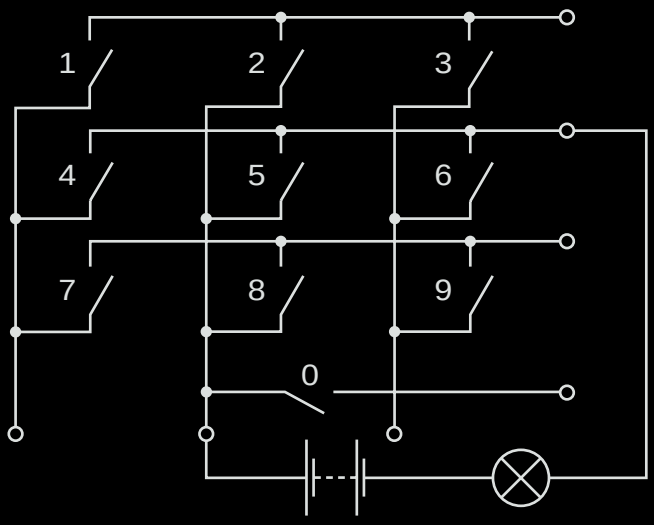
A difficult to magnetise and easy to demagnetise

B difficult to magnetise and retains magnetic strength

C easy to magnetise and retains magnetic strength

D easy to magnetise and easy to demagnetise

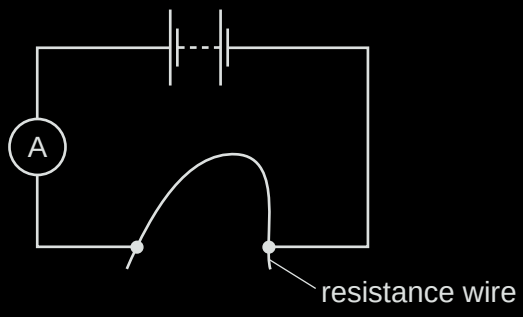
www.PapaCambridge.com



Which single switch can be pressed to make the lamp light?

A 0 **B** 1 **C** 5 **D** 6

27 A length of resistance wire is used as a resistor in a simple circuit.



Four separate changes are made to the wire.

Which change will **not** reduce the value of the resistance of the wire?

A It is covered in an insulating sleeve.

B Its cross-sectional area is increased.

C Its length is decreased.

D Its temperature is decreased.

28 Which quantity is measured in kilowatt-hours?

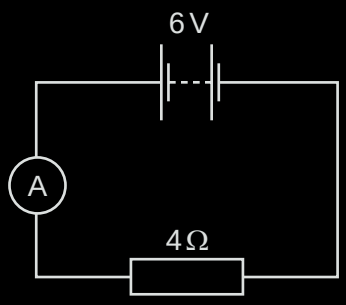
A charge

B current

C energy

D power

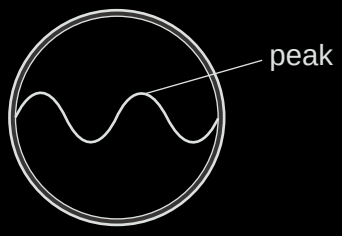
www.PapaCambridge.com



What is the reading on the ammeter?

A 0.67 A **B** 1.5 A **C** 10 A **D** 24 A

www.PapaCambridge.com

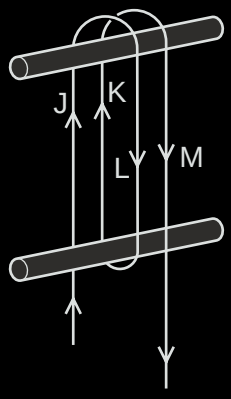


The speed of rotation of the generator is doubled.

What is the effect on the wave?

	number of peaks on the screen	amplitude of wave on the screen
A	doubled	doubled
B	doubled	same
C	same	doubled
D	same	same

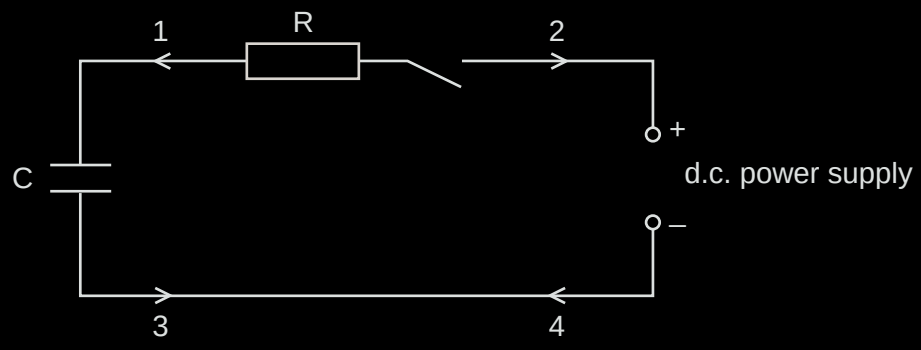
33 A long flexible wire is wrapped round two wooden pegs. A large current is passed in the direction shown.



Which two pairs of lengths of wire attract each other?

	first pair	second pair
A	J and K	K and M
B	J and K	L and M
C	J and L	K and M
D	J and L	L and M

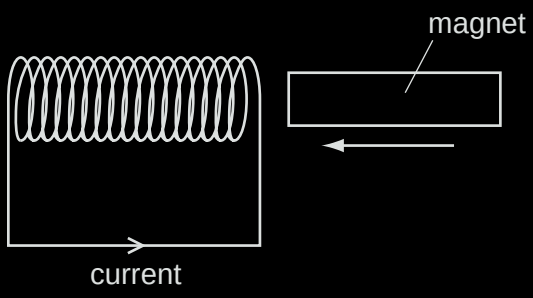
www.PapaCambridge.com



Which arrows show the direction of the conventional current when the capacitor is charging?

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

35 A magnet is pushed slowly into a coil and there is a current in the coil in the direction shown.

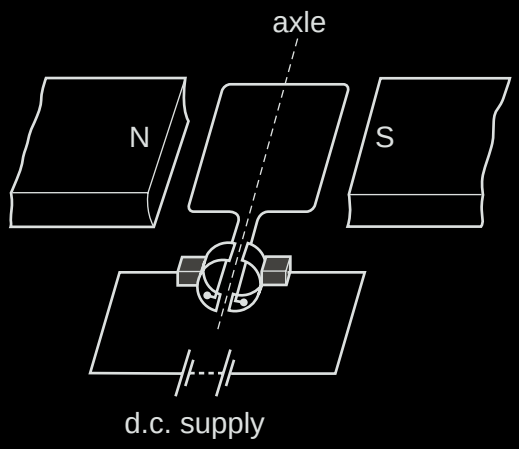


The magnet is then pulled out quickly from the same end of the coil.

What happens to the direction and the size of the current?

	direction	size
A	reversed	decreased
B	reversed	increased
C	unchanged	decreased
D	unchanged	increased

www.PapaCambridge.com



Why is a split-ring commutator used?

A to change the current direction in the coil as the coil passes the horizontal position

B to change the current direction in the coil as the coil passes the vertical position

C to change the current direction in the d.c. supply as the coil passes the horizontal position

D to change the current direction in the d.c. supply as the coil passes the vertical position

37 Which row is correct for fission and for fusion?

	fission of a nucleus	fusion
A	produces larger nuclei	is the energy source of a star
B	produces larger nuclei	is used to release energy in a power station
C	produces smaller nuclei	is the energy source of a star
D	produces smaller nuclei	is used to release energy in a power station

38 In one radioactive decay, radium (Ra) gives rise to radon (Rn) as shown.

$$^{226}_{88}\text{Ra} \rightarrow {}^{222}_{86}\text{Rn}$$

What particle is also produced?

A an alpha-particle

B a beta-particle

C both an alpha-particle and a beta-particle

D no particle but only gamma-rays

www.PapaCambridge.com

www.PapaCambridge.com

www.PapaCambridge.com

www.PapaCambridge.com